

# CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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**Abstract.** Student counseling in higher education draws on heterogeneous institutional knowledge, including relational curricula, cohort-specific tuition schedules, statutory exemption criteria, scholarship policies, and narrative administrative regulations. This setting poses a challenge for uniform retrieval and reasoning pipelines, as different forms of institutional knowledge may benefit from different representations, retrieval mechanisms, and processing strategies. We present CTU-Chat, a centralized supervisor-routed, domain-specialized multi-agent system for Vietnamese university counseling. The supervisor directs each query to one of four specialists, while each specialist is assigned an appropriate knowledge source, retrieval path, and set of tools. Relational academic information is handled through a knowledge graph; structured tuition and scholarship information is supported by dedicated lookup and calculation functions; and narrative regulations are retrieved through hybrid document search. Evaluation shows 97.0% raw structured-supervisor routing accuracy, while the proposed retrieval configuration reaches 0.7500 Hit@1 and 0.8354 MRR@10 on a redesigned 100-query held-out test set. These findings suggest that centralized coordination with domain-specific knowledge allocation is a promising approach for heterogeneous university counseling.

**Keywords:** Multi-Agent Systems · Domain-Specialized Agents · Centralized Supervisor Routing · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

## 1 Introduction

Can Tho University (CTU) is a key higher education institution in the Mekong Delta, with nearly 49,000 undergraduate students in the 2025–2026 academic year [1]. At an institution of this scale, advisory needs span a wide range of academic and administrative areas. Relevant regulations and policies may also change

over time. Prior research on automated advisory systems includes CAAS, which supports chat- and voice-based admission counseling for information-technology programs [10], and REBot, which focuses on academic-regulation advising through dense retrieval and category-guided graph reasoning [9]. Their reported evaluations illustrate the value of task-specific retrieval and processing strategies, but remain centered on individual advisory domains. This motivates CTU-Chat as a coordinated multi-domain architecture for heterogeneous university counseling.

University counseling presents challenges not only because it covers multiple domains, but also because institutional knowledge is represented in different forms. (1) **Academic curricula** contain hierarchical structures and prerequisite relationships; (2) **tuition schedules** are organized by admission cohort and field of study; (3) **tuition exemption policies** and (4) **scholarship policies** combine eligibility conditions with specific support rates and thresholds; while (5) **general administrative regulations** are presented in lengthy narrative documents. Therefore, no single representation or retrieval method is suitable for every domain, motivating representation-specific evidence acquisition in CTU-Chat.

RAG and GraphRAG augment LLMs with external documents and knowledge graphs [8,5,3], while agentic architectures support tool invocation and multi-agent coordination [20,18,16]. However, these approaches do not by themselves determine how heterogeneous institutional evidence should be represented, routed, and processed. Moreover, known limitations in language-model arithmetic [14] motivate deterministic calculation for rule-based financial queries.

These challenges highlights three open research gaps in automated university counseling: (1) limited domain-specialized multi-agent orchestration; (2) uniform representation of heterogeneous institutional knowledge; and (3) the lack of representation-aware evidence routing. These gaps are examined in greater detail in Section 2.

The objective of this study is to develop and evaluate CTU-Chat for heterogeneous university counseling. To address the identified gaps, we formulate three research questions:

- **RQ1 (Domain-Specialized Multi-Agent Architecture):** How effectively can a centralized supervisor coordinate domain-specialized agents with isolated knowledge spaces and constrained tool access for heterogeneous university counseling tasks?
- **RQ2 (Heterogeneous Knowledge Allocation & Modular Ablations):** What differences in context and answer quality are observed between the full heterogeneous configuration, uniform retrieval baselines, and its component ablations?
- **RQ3 (Representation-Aware Retrieval Effectiveness):** How does the complete representation-aware evidence configuration compare with sparse, dense, hybrid, and reranked baselines in retrieval effectiveness, both overall and across counseling domains?

We hypothesize that reliable university counseling is better supported by a supervisor-routed multi-agent architecture in which each specialist is restricted

to domain-appropriate knowledge representations, retrieval paths, and tools. To investigate this hypothesis and the research questions above, we propose CTU-Chat, which coordinates four domain-specialized agents for academic, financial, scholarship, and general administrative inquiries within a unified framework combining graph-based retrieval, structured data access, deterministic calculation, and hybrid document retrieval.

The remainder of this paper is organized as follows. **Section 2** reviews related work and summarizes the research gaps. **Section 3** presents the CTU-Chat architecture, evidence-acquisition methods, and domain-specialized agents. **Section 4** reports the experimental setup, results, and discussion. Finally, **Section 5** concludes the paper and outlines future work.

## 2 Related Work

### 2.1 Vietnamese NLP Foundations and Linguistic Specifics

Unlike whitespace-delimited languages, Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [11], with over 85% of lexical units forming multi-syllabic compound words (e.g., “*học phí*” [tuition fee]). While domain-adapted bi-encoders like PhoBERT capture semantic paraphrasing, Vietnamese higher education decrees rely heavily on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes. This dual characteristic motivates coupling syllable-aware dense embeddings with exact lexical token matching to mitigate both semantic drift and alphanumeric recall failures.

### 2.2 Sparse and Dense Retrieval

**Lexical Retrieval (BM25):** Sparse keyword search using BM25 [15] scores documents via term frequency (tf) and inverse document frequency (IDF) over an inverted index ( $k_1 = 1.5, b = 0.75$ ). BM25 provides reliable exact token matching for rigid alphanumeric identifiers (decision codes, cohort labels, course codes), but cannot capture semantic paraphrases.

**Dense Semantic Retrieval:** Dense retrieval systems [7,8] map queries and passages into embedding spaces using bi-encoders, computing relevance via vector similarity. While dense models capture paraphrastic intent, rare alphanumeric identifiers and out-of-vocabulary regulatory tokens can remain difficult and may introduce cross-cohort noise, motivating hybrid retrieval.

### 2.3 Hybrid Retrieval and Reciprocal Rank Fusion

To merge BM25 and dense candidates without score calibration, Reciprocal Rank Fusion (RRF) [2] sorts documents by reciprocal position:  $s_{\text{RRF}}(d) = \sum_{s \in \mathcal{M}} \frac{1}{k + r_s(d)}$  ( $k = 60$ ), while neural cross-encoders [12,19] jointly encode query–passage pairs for reranking. Adaptive-RAG further demonstrates that retrieval strategies can be selected according to question complexity [6]. These approaches motivate dynamic retrieval, but do not directly allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

## 2.4 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples  $(h, r, t)$  [13]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths escaping flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious or hallucinated relationships, especially when narrative documents lack explicit relational structure. Therefore, this work adopts selective rather than universal graph construction.

## 2.5 Agentic RAG and Multi-Agent Orchestration

ReAct demonstrates interleaved reasoning and action, AutoGen provides a framework for multi-agent conversation, and Toolformer studies when and how language models invoke external APIs [20,18,16]. For institutional counseling, these capabilities raise design risks when applied without explicit constraints: broad tool exposure can increase tool and parameter errors; peer-to-peer exchanges can increase coordination cost; a shared index can mix evidence across domains; and a common retrieval path can obscure opportunities for direct graph traversal or structured lookup. These risks motivate the bounded, supervisor-routed specialization evaluated in this work.

## 2.6 Educational and University Counseling Systems

Educational chatbots assist in advising and learning support [17], with recent systems like REBot [9] combining text retrieval with category-guided GraphRAG for academic regulations. CTU-CHAT departs from prior advisory systems across three key aspects: (1) *Task Scope*: while representative prior advisory systems often emphasize single-domain text-QA or focus primarily on narrative regulations, CTU-CHAT models 113 curricula as relational property graphs and incorporates deterministic arithmetic engines (`tin_h_toan_hoc_phi`, `tin_h_tien_hoc_bong`) for cohort-dependent tuition, exemptions, and scholarship thresholds; (2) *Multi-Agent Routing*: rather than a monolithic classifier-guided pipeline, a LangGraph supervisor dynamically routes queries to four bounded specialists with isolated evidence spaces and asymmetric tool privileges; and (3) *Metric Incommensurability*: prior benchmarks evaluate traditional classification accuracy and binary Q&A  $F_1$ -scores, precluding direct numerical comparison with CTU-CHAT, which evaluates end-to-end generative fidelity via Ragas and bounded multi-agent tool execution benchmarks.

## 2.7 Comparative Synthesis and Research Gaps

Synthesizing established retrieval and reasoning paradigms highlights clear operational trade-offs for university counseling [2,3,6,7,15,16,18,20]. While individual techniques resolve isolated needs—lexical matching for exact regulatory tokens, dense embeddings for semantic paraphrasing, and GraphRAG for relational

prerequisites—none simultaneously address the tripartite challenge of multi-domain institutional advising.

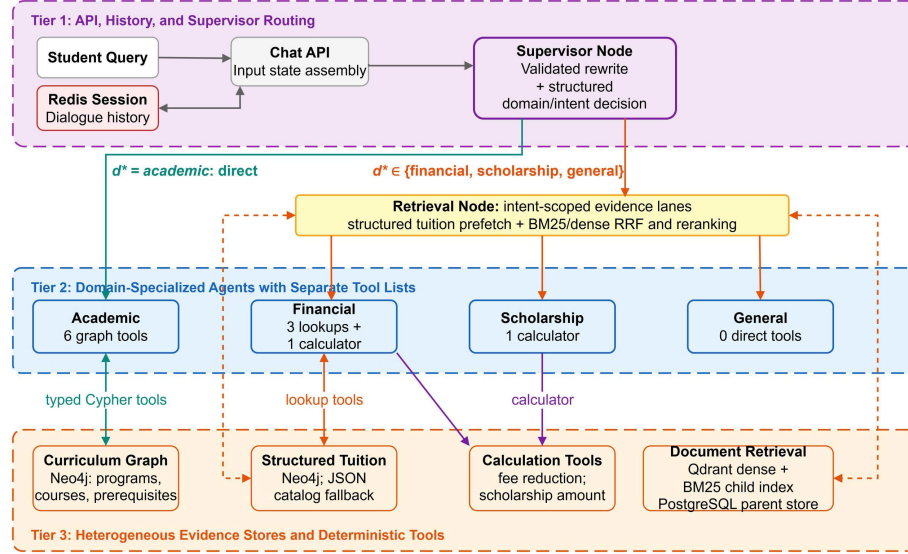
This analysis exposes three operational gaps in automated university counseling: (1) *Lack of bounded multi-agent orchestration*: monolithic pipelines and unconstrained agent meshes lack execution guardrails, frequently triggering tool mis-selection and parameter hallucination; (2) *Limitations of uniform representation*: forcing relational curricula, tabular tuition matrices, and narrative legal text into a single retrieval space can cause prerequisite fragmentation and semantic drift; and (3) *Absence of representation-aware routing*: dynamic steering mechanisms that allocate queries across isolated representation pathways while offloading statutory arithmetic to deterministic engines remain largely unaddressed in higher-education administrative counseling. These structural requirements directly motivate the centralized supervisor-routed architecture detailed in Section 3.

### 3 Proposed Model

This section presents the architecture, algorithms, and evidence-acquisition mechanisms of CTU-CHAT. The proposed model is organized around three design principles: *centralized agent coordination*, in which a contract-checked supervisor dispatches each query to one domain agent without peer-to-peer delegation; *heterogeneous knowledge allocation*, which assigns each information structure to a corresponding evidence representation; and *representation-aware evidence routing*, which selects either a direct graph-tool path or an intent-scoped retrieval path according to the assigned domain. Each *specialist* is configured with domain-specific prompts, a scoped evidence path, and a restricted tool registry.

#### 3.1 Overall Proposed Architecture

Figure 1 illustrates the system architecture of CTU-CHAT, structured across three coordinated tiers: (1) the *Interaction and Centralized Supervisor Tier*, (2) the *Domain-Specialized Multi-Agent Core Layer*, and (3) the *Heterogeneous Knowledge Storage and Tool Tier*.



**Fig. 1.** Implemented CTU-CHAT architecture. Redis history enters the graph state; the supervisor’s structured decision is contract-checked and may be repaired before conditional dispatch (repair is internal, not a separate node). Academic queries enter the graph-enabled agent, whereas other domains pass through intent-scoped retrieval. Solid and dashed links denote direct tool access and retrieval-node evidence acquisition, respectively.

**Application-Level Specialization of LangGraph StateGraph:** CTU-CHAT specializes LangGraph through typed state, an intent-owner contract, isolated paths, and partitioned tool registries. The contract checks structured supervisor output before dispatch; specialists then operate without peer delegation and see only their assigned tools.

- **Node-scoped state updates:** `AgentState` retains the validated rewrite, raw and repaired routes, repair reason, retrieved evidence, and response. The supervisor writes routing fields, retrieval writes `context` and `retrieval_instruction`, and the specialist writes `response`, making route repair traceable.
- **Two-Tier Conditional Routing Edges:** Instead of letting agents dynamically negotiate execution flow, the orchestration topology is governed by deterministic conditional functions:

$$\text{Edge}_{\text{sup}}(\mathcal{S}) = \begin{cases} \text{'academic\_agent'}, & \text{if } \mathcal{S}[\text{'next\_agent'}] = \text{'academic'} \\ \text{'retrieval'}, & \text{otherwise,} \end{cases} \quad (1)$$

bypassing document retrieval for queries assigned to the academic agent. At the retrieval exit,  $\text{Edge}_{\text{ret}}(\mathcal{S})$  dispatches the retrieved context to the selected financial, scholarship, or general agent.

- **Acyclic outer orchestration:** The StateGraph has no peer-to-peer agent edges. Each selected domain-specialized agent returns to END, so the outer routing path cannot cycle. ReAct tool interactions inside an agent remain distinct from this outer graph topology.
- **Domain-specialized agents with asymmetric tools:** Academic, financial, and scholarship agents receive scoped tool lists; the general agent has no direct tool. A prompt gate requests one call only when required parameters are valid. If a clearly tool-backed query omits it, one bounded retry names the required tool; missing parameters instead trigger clarification.

### 3.2 Neo4j Knowledge Graph Specification and Cypher Tool Mechanics

**Schema defined by the ingestion layer:** The Neo4j graph models relational dependencies across academic counseling using a simplified schema, comprising over 1,500 entity nodes and 3,800 relational edges across 113 undergraduate curricula and statutory fee schedules. It incorporates 10 node labels (7 for curriculum: Program, Course, Block, etc.; and 3 for tuition: TuitionFee, TuitionPolicy, ExemptionBasisRate) interconnected via 13 relationship types (HAS\_BLOCK, CONTAINS, REQUIRES, PARALLEL\_WITH, HAS\_PLO, HAS\_PEO, REALIZED\_BY, HAS\_TRACK, BELONGS\_TO\_TRACK, HAS\_JOB\_POSITION, HAS\_TUITION, GOVERNED\_BY, and HAS\_EXEMPTION\_BASIS). Narrative regulations remain in the document store unless linked to structured entities.

**Cypher tool mechanics:** The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies the tool arguments. The tool normalizes identifiers or names and calls a fixed, parameterized Cypher query in the graph service. The tools implement three query classes:

- *Attribute lookup and search:* `tra_cuu_nganh` and `tim_nganh` match program codes or normalized program, unit, and course names; `so_sanh_nganh` applies the same lookup before comparing two programs.
- *Structural traversal:* `xem_chuoitien_quyet` follows REQUIRES paths up to ten hops and returns direct PARALLEL\_WITH courses. `mon_chung_giua_nganh` computes a course intersection across two program–block paths, and `tim_nganh_co_mon` traverses those paths in reverse.
- *Structured financial lookup:* parameterized queries match TuitionFee, TuitionPolicy, and ExemptionBasisRate properties and their program relationships. This graph lookup is separate from the BM25–dense hybrid document retrieval path used for narrative regulations.

This design binds agent actions directly to deterministic graph operations: the LLM only selects the appropriate tool and supplies its arguments, while the query topology remains fixed and parameterized within the graph service. As a result, the multi-agent workflow reliably incorporates graph evidence while mitigating the syntactic and semantic fragility typically associated with unconstrained Cypher generation.

### 3.3 Algorithmic Foundations: Routing and Deterministic Arithmetic

Algorithm 1 specifies the supervisor-routing and evidence-path dispatch procedure.

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**Algorithm 1:** Supervisor Routing and Conditional Evidence-Path Dispatch

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**Input:** Student query  $q$ , Dialogue history  $\mathcal{H}$ , Shared state  $\mathcal{S}$ , Ownership contract  $\mathcal{P}$

**Output:** State updates and next graph node

```

1  $q_{\text{standalone}} \leftarrow q$ ;
2  $q_{\text{prev}} \leftarrow \text{MostRecentUserQuery}(\mathcal{H})$ ;
3 if ShouldRewriteQuery( $q, q_{\text{prev}}$ ) then
4    $q_{\text{cand}} \leftarrow \text{FastLLM.Rewrite}(q, q_{\text{prev}})$ ;
5   if ValidateRewrite( $q, q_{\text{cand}}, q_{\text{prev}}$ ) then
6      $q_{\text{standalone}} \leftarrow q_{\text{cand}}$ ;
7  $\langle d_{\text{raw}}, \iota_{\text{raw}} \rangle \leftarrow \text{StructuredSupervisor}(q)$ ;
8 if structured classification fails then
9    $d_{\text{raw}} \leftarrow \text{'general'}$ ;  $\iota_{\text{raw}} \leftarrow \text{'other'}$ ;
10  $\langle d^*, \iota^*, \rho \rangle \leftarrow \text{RepairRoute}(q, d_{\text{raw}}, \iota_{\text{raw}}, \mathcal{P})$ ;
11 Write  $q_{\text{standalone}}, d_{\text{raw}}, \iota_{\text{raw}}, d^*, \iota^*, \rho$  to  $\mathcal{S}$ ;
12 if  $d^* = \text{'academic'}$  then
13    $v_{\text{next}} \leftarrow \text{'academic\_agent'}$ ;
14 else
15   if  $d^* = \text{'financial'}$  then
16     Prefetch tuition evidence from Neo4j, else JSON;
17   Build retrieval lanes from  $\iota^*$  and retrieve evidence;
18    $v_{\text{next}} \leftarrow d^*_{\text{-agent}}$ ;
19 return  $\langle \mathcal{S}, v_{\text{next}} \rangle$ ;

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Rewrites require validation, and structured classification falls back to *general/other*. **RepairRoute** preserves valid predictions but corrects strong rule conflicts or incompatible agent–intent pairs under  $\mathcal{P}$ . Conditional edges send academic queries to the graph agent and other domains through retrieval.

**Deterministic Tuition Reduction Arithmetic:** For fee-reduction requests, CTU-CHAT augments the LLM with the external deterministic tool `tinh.toan.hoc.phi`. The LLM interprets the request and supplies three parameters—the actual per-credit tuition  $T$ , the per-credit exemption basis  $B$ , and the percentage entitlement  $p$ —but does not perform the arithmetic through free-form text generation. The external tool validates the percentage range and applies the fixed calculation

$$D = B \frac{p}{100}, \quad P = \max(0, T - D). \quad (2)$$

where  $D$  is the reduction per credit and  $P$  is the remaining amount per credit. Monetary inputs are expected in VND per credit. The tool enforces  $0 \leq p \leq 100$  and clamps  $P$  at zero; values outside the percentage range return an error.

Rather than a new general-purpose algorithm, the contribution lies in composing validated rewriting, constrained routing, representation-specific dispatch, and deterministic arithmetic within the multi-agent workflow.

### 3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

This layer implements *heterogeneous knowledge allocation*: rather than forcing all institutional knowledge into a uniform document index, CTU-CHAT preserves relational curriculum knowledge in Neo4j, accesses cohort-dependent tuition data through structured lookup, and retains narrative regulations and scholarship policies in document retrieval.

Table 1 makes this policy explicit by separating the physical evidence representation from the mechanism used to access it at runtime.

**Table 1.** Decision criteria mapping query characteristics to evidence representation mechanisms.

| Query characteristic         | Evidence path         | Rationale                  |
|------------------------------|-----------------------|----------------------------|
| Program/course attributes    | Neo4j property lookup | Exact or normalized match  |
| Prerequisites/shared courses | Neo4j traversal       | Stored graph relationships |
| Regulation identifiers       | BM25 child index      | Exact token match          |
| Paraphrased regulations      | Hybrid RRF + reranker | Lexical + semantic signals |
| Fee/scholarship calculation  | Python tool           | Fixed formula outside LLM  |

At runtime, academic queries enter the graph agent directly; other domains enter retrieval. Financial retrieval first attempts Neo4j tuition lookup with JSON fallback. The repaired intent then selects metadata-filtered document lanes, and packed evidence is passed only to the selected specialist.

**Hybrid Document Retrieval Path.** For narrative regulations and scholarship criteria, evidence is acquired through the following pipeline:

1. *Child-level candidate generation.* A BM25 index searches child chunks, while Qdrant performs dense retrieval over their 768-dimensional embeddings.
2. *Parent resolution.* Parent identifiers associated with the retrieved child chunks are used to load the corresponding parent sections from PostgreSQL.
3. *Reciprocal Rank Fusion.* The dense- and BM25-derived parent rankings are combined as

$$s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} \frac{1}{k + r_s(d)}, \quad k = 60.$$

4. *Cross-encoder reranking.* The fused parent candidates are reranked using BAAI/bge-reranker-v2-m3, and up to six parent sections are supplied to the selected domain agent.

Parent sections use a 2,800-character target with 100-character overlap; child chunks use a 400-character target with 50-character overlap.

## 4 Experimental Results

This section presents the empirical evaluation of CTU-CHAT.

### 4.1 Experimental Benchmark and Protocols

We evaluate CTU-CHAT on 200 administrative queries constructed through a document-grounded semi-automated protocol over official Can Tho University (CTU) statutes. Candidate queries with diverse conversational forms were generated from statutory articles and reviewed for ground-truth intents, passages, required facts, and answers. The 100-query development split supports index calibration, prompt engineering, and failure analysis. The separately constructed 100-query held-out set follows a predefined complexity distribution of 40 direct single-hop, 20 within-domain multi-hop, 20 cross-domain, 10 entity-comparison, 5 temporal, and 5 adversarial queries. It includes 51 formal and 49 colloquial formulations; 56 queries have one gold source and 44 require multiple gold sources.

An audit using entity inventories separate from the development split found no exact cross-split duplicates and a maximum 5-gram Jaccard similarity of 0.1875. Gold sources and required facts were reviewed against 244 institutional documents and 559 structured tuition records. The held-out set contains 19 Academic, 28 Financial, 17 Scholarship, and 16 General single-domain queries; the remaining 20 combine evidence across domains and are analyzed separately. Tables 2 and 4 report results on this held-out set.

**Evaluated Snapshot and Infrastructure:** Reported results use a fixed Gemini 2.5 Flash Lite snapshot through Google Cloud Vertex AI with greedy decoding ( $T = 0.0$ ); later deployment-model substitutions are not evaluated. Each query was run  $R = 3$  times, yielding 2,100 generation and Ragas turns (7 configurations  $\times$  100 queries  $\times$  3 runs). Neural reranking used an NVIDIA RTX 1060 GPU (4 GB VRAM). Neo4j 5 Enterprise with APOC stored relational curricula; PostgreSQL stored parent chunks (2,800 characters, 100 overlap); and Qdrant indexed child chunks (400 characters, 50 overlap) using a 768-dimensional vietnamese-bi-encoder. The system used `bge-reranker-v2-m3`, LangGraph StateGraph, and Redis 7.

### 4.2 Evaluation Metrics

**Evaluation Metrics:** Scenario 1 reports source-document-level Hit@ $k$  ( $k \in \{1, 3\}$ ), Precision@5, Recall@5, MRR@10, and domain-level Hit@1. Scenario 3 measures routing accuracy, Macro-F1, intent accuracy, tool selection and argument Exact Match (EM), end-to-end pass rate, adversarial tool gating, and safe behavior under `max_tool_calls = 1` and a 60-s timeout.

Scenario 2 evaluates Top-7 synthesis using Ragas [4]: Answer Relevancy (AR), Context Recall (CR), Context Precision (CP), and Answer Correctness (AC). Factual Exact Match (Fact EM) measures strict numerical tuition-fee and credit agreement, while Source Recall measures gold-source coverage and Source Average Precision (Source AP) evaluates ranked, verified regulatory citations. To limit LLM-judge bias [21,4], evaluation conditions reference-grounded metrics on gold documents, uses greedy decoding over three runs, and includes programmatic Fact EM for tuition arithmetic.

### 4.3 Experimental Scenarios and Benchmark Results

We evaluate CTU-CHAT through retrieval component stacking (RQ3), end-to-end generation and ablation (RQ2), and multi-agent coordination and tool reliability (RQ1).

**Scenario 1 — Retrieval Component Stacking (E1–E5):** Scenario 1 compares five cumulative retrieval configurations, from sparse and dense baselines to hybrid fusion, neural reranking, and the proposed intent-guided evidence stack, at aggregate and complexity-stratified levels on the 100-query held-out set.

Table 2 shows that E5 obtains the highest aggregate retrieval scores: 0.7500 Hit@1, 0.9100 Hit@3, 0.2380 Precision@5, 0.8458 Recall@5, and 0.8354 MRR@10. Relative to E4, E5 increases Hit@1 by 7.0 percentage points and MRR@10 by 0.0450; relative to E1, its Hit@1 is 22.0 points higher. Mean retrieval latency increases from 15.62 s for E4 to 17.09 s for E5. Because E5 jointly adds intent governance, graph grounding, and parent–child mapping, this contrast compares the integrated bundle rather than any isolated component.

**Table 2.** Scenario 1: Progressive retrieval component stacking on the redesigned Held-Out Test Set ( $N = 100$ ).

| Retrieval Configuration                | Hit@1         | Hit@3         | P@5           | Recall@5      | MRR@10        | Latency (ms) |
|----------------------------------------|---------------|---------------|---------------|---------------|---------------|--------------|
| E1: BM25 Lexical Baseline              | 0.5300        | 0.7600        | 0.1980        | 0.7133        | 0.6661        | <b>9.79</b>  |
| E2: Dense Vector Semantic              | 0.4200        | 0.6300        | 0.1540        | 0.5542        | 0.5223        | 50.80        |
| E3: Hybrid RRF (BM25 + Dense)          | 0.5000        | 0.7800        | 0.2040        | 0.7308        | 0.6568        | 53.10        |
| E4: Hybrid + Neural Reranker           | 0.6800        | 0.9000        | 0.2320        | 0.8275        | 0.7904        | 15,624.33    |
| <b>E5: CTU-Chat Proposed Retrieval</b> | <b>0.7500</b> | <b>0.9100</b> | <b>0.2380</b> | <b>0.8458</b> | <b>0.8354</b> | 17,094.94    |

E5’s largest Hit@1 differences over E4 occur for entity-comparison and temporal queries (20 percentage points each) and cross-domain queries (15 points). Hit@1 is unchanged for multi-hop and adversarial queries, although E5 improves Recall@5 on both tiers. For cross-domain queries, E5 improves Hit@1 but has slightly lower Recall@5 (0.7750 vs. 0.7833).

**Scenario 2 — End-to-End QA Generation and Modular Ablation (T1–T7):** Scenario 2 compares the full system (T4) with three single-agent RAG baselines (T1–T3) and three ablations (T5–T7) over 2,100 evaluation turns. As shown in Table 4, T4 exceeds BM25 on all seven metrics, including a 7.13-percentage-point difference in Fact EM (65.29% vs. 58.16%). T4 records the

**Table 3.** E4–E5 retrieval results by held-out query complexity.

| Complexity Tier         | $N$ | Hit@1  |               | Recall@5      |               |
|-------------------------|-----|--------|---------------|---------------|---------------|
|                         |     | E4     | E5            | E4            | E5            |
| Direct single-hop       | 40  | 0.6500 | <b>0.6750</b> | 0.9000        | <b>0.9250</b> |
| Within-domain multi-hop | 20  | 0.7000 | 0.7000        | 0.7167        | <b>0.7417</b> |
| Cross-domain            | 20  | 0.7000 | <b>0.8500</b> | <b>0.7833</b> | 0.7750        |
| Entity comparison       | 10  | 0.8000 | <b>1.0000</b> | 0.9500        | 0.9500        |
| Temporal                | 5   | 0.8000 | <b>1.0000</b> | 1.0000        | 1.0000        |
| Adversarial             | 5   | 0.4000 | 0.4000        | 0.4500        | <b>0.5500</b> |

**Table 4.** Scenario 2: End-to-end QA generation and modular ablation on the redesigned Held-Out Test Set ( $N = 100$ ,  $R = 3$ , 2,100 total evaluation turns, Top-7 context).

| Cfg.      | CP            | CR            | AR            | AC            | Fact EM       | Src. R        | Src. AP       |
|-----------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| T1        | 0.2576        | 0.5411        | 0.3961        | 0.4758        | 0.5816        | 0.7308        | 0.5498        |
| T2        | 0.1664        | 0.2682        | 0.2281        | 0.3607        | 0.5443        | 0.5642        | 0.4240        |
| T3        | 0.2110        | 0.4843        | 0.3618        | 0.4631        | 0.5965        | 0.7608        | 0.5469        |
| <b>T4</b> | <b>0.3833</b> | <b>0.6296</b> | <b>0.4454</b> | <b>0.5080</b> | <b>0.6529</b> | <b>0.8408</b> | <b>0.7313</b> |
| T5        | 0.2255        | 0.4617        | 0.2996        | 0.4400        | 0.5857        | 0.7192        | 0.6104        |
| T6        | 0.3761        | 0.5979        | <b>0.4639</b> | <b>0.5108</b> | 0.6271        | <b>0.8408</b> | 0.7058        |
| T7        | <b>0.3862</b> | <b>0.6572</b> | 0.4235        | 0.4992        | 0.6384        | 0.8208        | 0.7238        |

*Configurations:* T1: BM25; T2: dense; T3: hybrid RRF; T4: full system; T5: without cross-reranker; T6: without Neo4j grounding; T7: without subspace governance. *Metrics:* Fact EM: Factual Exact Match on tuition/credits; CP: Context Precision; CR: Context Recall; AR: Answer Relevancy; AC: Answer Correctness; Src. R: Source Recall; Src. AP: Source Average Precision. Ragas values are means over 300 query–repetition observations per configuration; corresponding standard deviations and bootstrap confidence intervals are retained in the experiment artifacts. All metrics are normalized to  $[0, 1]$ ; higher is better.

highest Fact EM and Source AP and ties T6 on Source Recall; T6 leads Answer Relevancy and Answer Correctness, whereas T7 leads Context Precision and Context Recall. The differences are therefore metric-specific.

### Scenario 3 — Multi-Agent Routing, Tool Reliability, and Robustness:

Scenario 3 evaluates coordination, tools, and bounded safety over 550 runs: supervisor routing (100 rule-based and 300 LLM runs), three specialist tools (90 runs), and 20 adversarial cases repeated three times (60 runs; Table 5). Panel A scores raw supervisor output before route repair; Panels B–C test isolated one-decision gates, not the full ReAct retry path.

## 4.4 Discussion

**Answering RQ1:** Before production route repair, the evaluated raw structured supervisor achieves 97.0% routing accuracy, 97.8% Macro-F1, and 95.0% intent

**Table 5.** Scenario 3: Multi-agent routing, specialist tool execution reliability, and robustness under a maximum of one tool call per decision (`max_tool_calls` = 1).

| Panel A: Raw Supervisor Routing and Specialist Selection ( $N = 100$ queries; Rule: 1 rep / LLM: 3 reps, $N = 300$ ) |              |              |                |              |               |                     |                    |
|----------------------------------------------------------------------------------------------------------------------|--------------|--------------|----------------|--------------|---------------|---------------------|--------------------|
| Router Variant                                                                                                       | Runs ( $N$ ) | Agent Acc.   | Macro-F1       | Intent Acc.  | Bounded       | Routing Paradigm    |                    |
| Rule-Based Router                                                                                                    | 100          | 85.0%        | 69.2%          | 69.0%        | 100.0%        | Heuristic Regex     |                    |
| Raw LLM Supervisor (CTU-CHAT)                                                                                        | 300          | <b>97.0%</b> | <b>97.8%</b>   | <b>95.0%</b> | <b>100.0%</b> | Structured Pydantic |                    |
| Panel B: Specialist Tool Execution Reliability ( $N = 90$ runs across 30 test cases, 3 repeated runs)                |              |              |                |              |               |                     |                    |
| Specialist Tool Function                                                                                             | Runs ( $N$ ) | Selection    | Arg. EM        | Result Acc.  | E2E Pass      | Bounded             | Paradigm           |
| tuition.lookup                                                                                                       | 30           | 100.0%       | 100.0%         | 100.0%       | 100.0%        | 100.0%              | Structured Catalog |
| scholarship.calculation                                                                                              | 30           | 86.7%        | 76.7%          | 86.7%        | 76.7%         | 100.0%              | LLM Tool Selection |
| tuition_reduction.calculation                                                                                        | 30           | 100.0%       | 100.0%         | 100.0%       | 100.0%        | 100.0%              | Arithmetic Tool    |
| Panel C: Robustness and Bounded Failure Handling ( $N = 60$ runs under malformed/adversarial inputs)                 |              |              |                |              |               |                     |                    |
| Adversarial Gate                                                                                                     | Runs ( $N$ ) | Agent Acc.   | Tool Gate Safe | Result       | E2E Pass      | Bounded             | Notes              |
| Failure Gate Robustness                                                                                              | 60           | 100.0%       | 100.0%         | 98.3%        | 98.3%         | 100.0%              | –                  |

*Note:* `tuition_lookup` evaluates deterministic structured catalog queries; `scholarship` and `tuition_reduction` cases evaluate LLM-driven parameter extraction and external deterministic tool execution.

accuracy, compared with 85.0%, 69.2%, and 69.0% for rule-based routing. The deterministic tuition tools achieve 100.0% end-to-end pass rates, whereas scholarship calculation reaches 76.7% with 86.7% tool selection; all tools maintain 100.0% bounded completion. Adversarial tests obtain 100.0% tool-gating accuracy, 98.3% safe-result behavior, and 100.0% bounded completion. In a paired stress-test with 11 tools, single-agent and multi-agent end-to-end performance is 93.3% and 88.3%, respectively, without a statistically significant difference. After adding 16 synthetic distractor tools (27 tools total), monolithic tool selection decreases from 98.3% to 91.7%, whereas the domain-partitioned system maintains 100.0% (paired  $\Delta = +8.3$  percentage points, 95% bootstrap CI  $[+1.7, +15.0]$ ,  $n = 60$ ). These results suggest that partitioned tool access may reduce decision-space interference as the evaluated registry expands.

**Answering RQ2:** T4 exceeds the no-cross-reranker configuration T5 on every reported metric, supporting the contribution of cross-reranking in this setting. Comparisons with T6 and T7 are mixed: T4 leads T6 in Context Recall, Source AP, and Fact EM but trails it in Answer Relevancy and Answer Correctness and ties it on Source Recall. T4 leads T7 in Answer Relevancy, Answer Correctness, Source Recall, Source AP, and Fact EM but trails it in Context Precision and Context Recall. The ablations therefore indicate component-specific associations rather than uniform causal effects.

**Answering RQ3:** E4 improves both early-rank ordering and evidence coverage over E3, while E5 provides smaller aggregate gains. E5’s largest Hit@1 differences over E4 occur on entity-comparison, temporal, and cross-domain queries. Direct and within-domain multi-hop queries show smaller or no Hit@1 gains, and the adversarial tier remains difficult for both configurations. E5 also has slightly lower cross-domain Recall@5 despite its higher Hit@1. Because E5 changes three components jointly, these results do not isolate their individual causal effects.

#### 4.5 Threats to Validity and Limitations

The evaluation is limited to CTU and 100 held-out queries; the temporal and adversarial tiers contain only five each. Domain analysis covers 80 single-domain queries and excludes 20 cross-domain cases. Using Gemini 2.5 Flash Lite as generator and judge may introduce bias despite grounding and repeated deterministic runs; judges may also penalize valid abstentions. Scenario 3 isolates raw routing and one-decision gates rather than the production repair/retry path. Curriculum graphs use offline schemas, while live Neo4j/Qdrant fault injection and multi-institutional validation remain future work.

### 5 Conclusion and Future Work

This study investigated how centralized supervision and representation-specific evidence paths can support heterogeneous university counseling. Before production route repair, raw supervisor outputs matched domain specialists with 97.0% agent accuracy while maintaining bounded execution in the evaluated cases. On the redesigned 100-query held-out benchmark, the full retrieval configuration achieved the highest Hit@1 (0.7500), Hit@3 (0.9100), Precision@5 (0.2380), Recall@5 (0.8458), and MRR@10 (0.8354) among the evaluated configurations. Its largest Hit@1 differences relative to E4 occurred on entity-comparison, temporal, and cross-domain queries, although E4 retained slightly higher cross-domain Recall@5. The E4-to-E5 comparison reflects the integrated routing, graph-grounding, and parent-child mapping bundle rather than an isolated component effect. In end-to-end evaluation, the full system achieved the highest Fact EM and Source AP and tied for the highest Source Recall, but did not obtain the highest value on every LLM-judged metric. The ablation results therefore indicate metric-specific contributions rather than uniform superiority of the complete configuration.

Taken together, these findings suggest that no single representation or retrieval strategy dominates all university-counseling tasks. Relational curricula, structured financial information, and narrative regulations impose different evidence and reasoning requirements and are therefore better handled through separate but coordinated processing paths. In practice, the proposed architecture manages these paths within a unified deployment while restricting each specialist to its relevant knowledge sources and tools. The results support the integrated architecture in the evaluated setting, but do not establish that any individual component is universally optimal.

The evaluation remains limited to Can Tho University data, automated judging, offline knowledge schemas, small temporal and adversarial strata, and isolated robustness tests. The single-domain analysis also excludes the benchmark’s 20 cross-domain queries. Future work will expand the held-out benchmark using anonymized real-world student counseling logs, subject to appropriate consent and privacy safeguards, to improve coverage of naturally occurring intents and conversational variation. Further work will evaluate transferability across institutions, conduct user studies with students and advisors, perform controlled computational-efficiency evaluation and model compression, employ cross-model

evaluation to reduce judging bias, and compare CTU-CHAT with broader educational AI systems. It will also expand the production tool registry and evaluate matched single- and multi-agent configurations across increasing tool-set sizes, including production tools rather than synthetic distractors. Within the scope of this study, the central takeaway is not that one retrieval method should replace all others, but that institutional counseling may be better supported when evidence acquisition and execution constraints are aligned with the structure of each domain.

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